

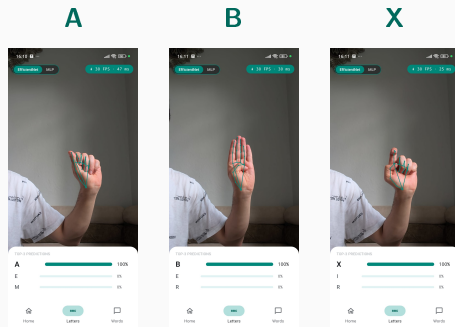
# On-Device American Sign Language Recognition

*Image- vs. Landmark-Based Approaches  
for Mobile Deployment*

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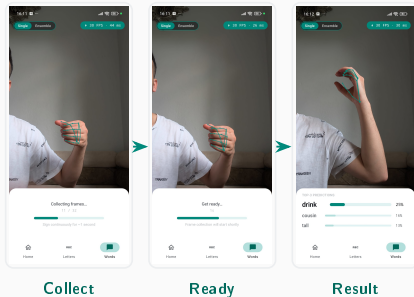


*Letters tab · landmark overlay*

## The Gap: Accurate on a server $\neq$ usable on a phone

- Sign language remains **underserved** by consumer technology
- Literature optimises offline **accuracy**, rarely **deployability**
- Two complementary tasks:
  - **Letters** — 29 static classes
  - **Words** — WLASL-100, temporal
- Hard constraint: **100 % on-device**, no server

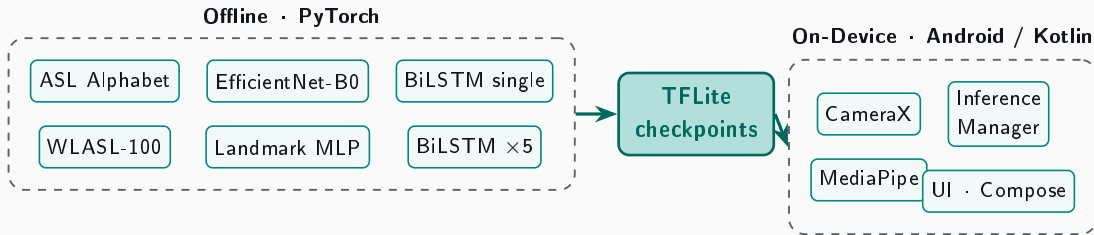
### Word recognition flow · “drink”



*32-frame window · BiLSTM latches top-1*

# System Architecture

One artefact bridges two worlds: the **TFLite checkpoint**



*Four models · two datasets · one Android app*

# Implementation

4 models · 2 datasets · 1 app

## Models · two tasks

### Letters task

- EfficientNet-B0 —  $224^2$  RGB, ImageNet pretrain
- Landmark MLP — 63-D MediaPipe pose

### Words task

- BiLSTM —  $128 \times 32$  two-hand landmark stream
- BiLSTM  $\times 5$  ens. — 5-seed logit averaging

## Datasets

- ASL Alphabet — 87,000 images · 29 classes
- WLASL-100 — 1,013  $\rightarrow$  1,465 clips  
yt-dlp recovery, +44 % over Kaggle baseline

## Ablation Technique

Forward-additive 9-step protocol on WLASL-100 val; each pillar isolated to measure its incremental gain.

### Four cumulative pillars

- Data engineering — yt-dlp recovery, re-balancing
- Extraction — two-hand pose, hand-crop norm.
- Training recipe — aug. + mixup + sampling
- Ensemble — 5-seed logit averaging

Data engineering =  
largest lever (+16 pp).

# Tests & Results

The offline ranking **inverts** on a budget phone

## Letters · Top-1 accuracy

|                 | Offline | TECNO   | Xiaomi  |
|-----------------|---------|---------|---------|
| EfficientNet-B0 | 100.0 % | 92.9 %  | 100.0 % |
| Landmark MLP    | 99.4 %  | 100.0 % | 100.0 % |

*Ships in 0.24 MB vs. 15.71 MB (EfficientNet).*

## Words · WLASL-100

|                 | Offline | Xiaomi |        |
|-----------------|---------|--------|--------|
|                 | Top-1   | Top-1  | Top-3  |
| BiLSTM (single) | 52.1 %  | 38.0 % | 71.0 % |
| BiLSTM ×5 ens.  | 56.2 %  | 49.0 % | 72.0 % |

*Ensemble clears 40 % bar; Top-3 ~72 % — usable UX.*

## Throughput · latency (ms) / FPS

|                 | Xiaomi 15T Pro | TECNO 20C |
|-----------------|----------------|-----------|
| EfficientNet-B0 | 35 / 30        | 203 / 21  |
| Landmark MLP    | 19 / 30        | 22 / 24   |
| BiLSTM (single) | 32 / 30        | 202 / 22  |
| BiLSTM ×5       | 80 / 30        | 665 / 17  |

*TECNO word models **not interactive**.*

The “right” model is  
device-conditional.